

Title

Sensorimotor circuit connectivity as a candidate biomarker for responsiveness to [sertraline](#) in obsessive–compulsive disorder

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Abstract

Predicting selective serotonin reuptake inhibitor (SSRI) response in obsessive–compulsive disorder (OCD) remains a clinical challenge. Converging evidence implicated that the sensorimotor circuit is linked to OCD-related sensory phenomena and repetitive motor rituals, and it is densely innervated by serotonergic projections, making it a plausible substrate of SSRI effects. We therefore hypothesized that baseline functional connectivity (FC) of this circuit could serve as a candidate neural marker of SSRI treatment response. In this exploratory single-site resting-state fMRI study, 54 drug-naïve patients with OCD and 39 matched healthy controls (HC) underwent scanning. Patients received sertraline for 12 weeks and were classified as responders (rOCD, $n = 33$) or non-responders (nOCD, $n = 21$) based on Yale-Brown Obsessive Compulsive Scale score reductions. Seed-based FC analysis of the sensorimotor circuit was conducted across the three groups. We observed that OCD patients exhibited abnormal FC primarily within the sensorimotor circuit and in its connections with the cerebellum. The rOCD group showed generally higher FC within the sensorimotor circuit than HC, whereas the nOCD group showed lower FC values. Cerebellar regions with altered connectivity included areas involved in sensorimotor processing and higher-level functions. In prediction analyses, the connectivity between right thalamus and cerebellar Crus I region achieved an AUC of 0.854 for distinguishing responders from non-responders under leave-one-out cross-validation. Moreover, FC-based models showed better predictive performance than clinical models. These findings suggest that baseline sensorimotor-network FC may serve as a candidate biomarker of sertraline response in OCD, pending validation in large, independent samples.

Introduction

Obsessive-compulsive disorder (OCD) is a chronic and disabling psychiatric disorder, characterized by distress-inducing intrusive thoughts (obsessions) and repetitive or ritualized behaviors (compulsions)[1]. Epidemiological studies indicate a global prevalence of approximately 2%-3%[2], with a lifetime prevalence in China reaching 2.4%[3]. Affected individuals experience significant functional impairment[4] and impose substantial socioeconomic burdens[5]. However, only approximately half of patients respond to first-line pharmacological strategies[6]—selective serotonin reuptake inhibitors (SSRIs)[7], and reliance on post hoc identification through sequential medication trials delays effective intervention and exacerbates disability burden. Although some clinical features—such as poor insight, religious or hoarding symptoms, and high comorbidity—have been linked to poor response[8], they offer limited predictive power and lack mechanistic specificity, highlighting the need for biomarkers that can prospectively distinguish SSRI responders from non-responders.

Pharmacological strategies for OCD are limited by an incomplete understanding of its pathological mechanisms. The most extensively validated dysfunctional network in OCD is the cortico-striato-thalamo-cortical (CSTC) loop, which governs processes such as habitual behavior, emotional regulation, and the integration of sensory and motor information[9,10]. The CSTC loop is composed of parallel yet partially segregated circuits, including the sensorimotor circuit that primarily connects the premotor cortex, putamen, and thalamus[11,12]. Although historically overlooked, the sensorimotor circuit was recently demonstrated by a mega-analysis from the Enhancing Neuro-Imaging and Genetics through Meta-Analysis (ENIGMA) OCD consortium to exhibit widespread hypoconnectivity across 28 cohorts[13]. Notably, dysfunction of the sensorimotor circuit is closely associated with several core clinical features of OCD, such as habitual behaviors, automatic responses and sensory phenomena (SP)[12]. In addition, evidence from neuropsychological research suggests that impaired response inhibition—a potential endophenotypic marker for OCD—engages the cortical and subcortical regions within the sensorimotor circuit[14-17], highlighting a more central role for sensorimotor circuit in the neurobiology of OCD.

Moreover, the sensorimotor circuit is structurally connected to the serotonergic system and may serve as a functional substrate for SSRI modulation. The raphe nuclei (RNi)—the primary source of 5-hydroxytryptamine (5-HT)—project densely to key nodes of the sensorimotor circuit, including thalamus and sensory cortices[18,19]. In addition, preclinical studies revealed that agonists of the 5-HT_{1B} receptor—a

presynaptic autoreceptor on serotonergic neurons—could lead to increased perseverative behaviors and impairments in sensorimotor gating in rodents, which could be attenuated by chronic SSRI treatment[20,21]. A similar correlation between 5-HT_{1B} receptor and dysfunctional sensorimotor gating was also observed in OCD patients using positron emission tomography (PET)[22]. Moreover, converging evidence further indicates that SSRIs can broaden network-level plasticity in circuits encompassing the sensorimotor circuit, enabling long-term potentiation (LTP)-like plasticity and longer-lasting responsiveness to neuromodulatory inputs[23]. Taken together, the evidence positions the sensorimotor circuit as a mechanistically plausible substrate through which serotonergic modulation may influence treatment outcomes.

Recent studies have used indices derived from the resting-state functional magnetic resonance imaging (rs-fMRI), such as functional connectivity (FC), [in an attempt to](#) predict SSRI response in OCD, with encouraging findings reported[24,25]. However, the selected seed regions either related to serotonergic pharmacology, such as RNI[24], or implicated in the putative mechanisms of OCD alone, such as cerebellum[25], limiting disorder specificity and mechanistic interpretability. To address this gap, we focus on the sensorimotor circuit as a prespecified seed region, given its position at the intersection of core OCD pathophysiology and the pharmacological effects of SSRIs. Additionally, without imposing seeds, whole-brain rs-fMRI studies have converged on the sensorimotor circuit as a prominent predictor of SSRI response[26,27]. For example, voxel-mirrored homotopic connectivity (VMHC) in the postcentral gyrus was found to be associated with SSRI efficacy in medication-free OCD patients[26]. Together, this evidence supports the sensorimotor circuit as a candidate that can offer disorder-relevant, mechanistically interpretable predictive value.

To address these limitations, we employed a seed-based FC approach to examine alterations of the sensorimotor circuit in drug-naïve OCD patients. [In this exploratory study, we first characterized baseline FC differences between sertraline responders \(rOCD\) and non-responders \(nOCD\), and compared both groups with healthy controls \(HC\). We then explored whether baseline FC values showed potential to predict sertraline treatment response using binary logistic regression and cross-validation.](#) We hypothesized that sensorimotor connectivity at baseline would differentiate rOCD from nOCD and serve as a candidate biomarker for [sertraline](#) response.

Materials and Methods

Participants and research procedure

A total of 59 drug-naïve OCD outpatients were recruited from Shanghai Mental

Health Center (SMHC) from September 2020 to August 2023. Key inclusion criteria were: age 18–65 years, DSM-5 diagnosis of OCD confirmed by psychiatrists[28], Yale–Brown Obsessive Compulsive Scale (Y-BOCS) score ≥ 20 [29], and no psychotropic medication or psychotherapy for OCD in the past 6 months. Forty-six age-, sex- and education level-matched HC were recruited through local advertisements. Patients whose primary psychiatric diagnosis was not OCD were excluded. Comorbid psychiatric disorders were screened using the Mini-International Neuropsychiatric Interview (M.I.N.I.)[30], and supplemented by patient self-report of lifetime diagnoses. Further inclusion/exclusion details are provided in Supplementary Methods.

Data were drawn from an existing clinical cohort (NCT04539951) described in detail elsewhere[31]. All participants underwent rs-fMRI scanning and provided demographic data at baseline. OCD patients subsequently received sertraline, initially at 50 mg/d, with weekly 50 mg/d increments up to the maximum recommended dosage (200 mg/d) or the maximum tolerated dosage (<200 mg/d). Final sertraline doses were extracted from clinical records. Dose information was missing for 8 of 54 patients due to incomplete documentation. Among the remaining 46 patients with available data, all received a final sertraline dose of at least 150 mg/day (Supplementary Figure 1). Patients were followed up at 2, 4, 8, and 12 weeks after treatment initiation, and medication adherence was assessed at each visit based on patient self-report. Patients who reported poor adherence or discontinuation of sertraline were excluded. OCD symptom severities were assessed using the Y-BOCS at baseline and after 12 weeks of treatment, and baseline depressive and anxiety symptoms were assessed with the Beck Depression Inventory (BDI)[32] and the Beck Anxiety Inventory (BAI) respectively[33]. Treatment response (rOCD) was defined as a $\geq 35\%$ reduction from baseline in Y-BOCS scores following treatment, whereas non-response (nOCD) was defined as a $<35\%$ reduction[34].

The study was approved by the Research Ethics Committee at Shanghai Mental Health Center, Shanghai Jiao Tong University School of Medicine. Participants provided written informed consent before enrollment.

Image acquisition

MRI data was acquired using a Siemens 3T Verio scanner at SMHC. Participants were instructed to relax, remain awake, keep their eyes closed and refrain from active thinking. Foam pads and earplugs were provided to reduce head motions and scanner noise. Anatomical images were acquired using a three-dimensional magnetization-prepared rapid acquisition gradient echo (3D MPRAGE) sequence, and resting-state

fMRI data using a standard gradient echo planar imaging sequence. [Full acquisition parameters are provided in Supplementary Methods.](#)

Data preprocessing

Preprocessing and quality assurance were performed by CONN Toolbox (version 22.v)[35], based on SPM version 12 (SPM12)[36], following standard procedures for realignment, normalization to MNI space, spatial smoothing, denoising (component-based correction and motion regression), and temporal band-pass filtering. [Additional preprocessing and quality-control details are provided in Supplementary Methods.](#) In total, 4 patients in the nOCD group, 1 patient in the rOCD group and 7 HC were excluded due to excessive head motion, leaving a final sample of 21 patients in the nOCD group, 33 patients in the rOCD group and 39 HC.

Functional connectivity analysis

FC analyses were conducted using CONN Toolbox. A total of 28 region of interests (ROI) were delineated using a combination of atlases: 24 ROIs matched to sensorimotor cortices from the 200-parcel Kong 2022 Atlas[37], and [4 bilateral subcortical ROIs from the Harvard-Oxford Atlas included the left/right thalamus and putamen](#)[38], resulting in 14 ROIs per hemisphere. Seed-based whole-brain FC maps were generated by computing Pearson correlation coefficients between the mean time series of each seed ROI and the time series of all other brain voxels. These correlation values were then transformed to Fisher's z-scores for subsequent group-level analyses. A one-way analysis of variance (ANOVA) was performed for each voxel to examine differences in FC among the three groups (nOCD, rOCD, HC) using a general linear model, with the sex, age and years of education as covariates. To determine statistically significant clusters of altered connectivity, we applied a two-step thresholding approach: a voxel-level threshold of $p < 0.001$, followed by a cluster-level family-wise error (FWE) correction at $p < 0.05$ (two-tailed).

Predictive modeling and validation

[Each candidate predictor, including FC and clinical features, was tested with univariate logistic regression and evaluated by leave-one-out cross-validation \(LOOCV\).](#) Within each fold, we adjusted FC values for clinical covariates—sex, age, education years, illness duration, baseline Y-BOCS, BDI and BAI—in the training set only, and the resulting coefficients were applied to the held-out case. [Predictive performance was summarized using effect-size and discrimination metrics, with threshold-based indices derived from the Youden index in the training set. For each](#)

FC, a null distribution was obtained by repeating the entire LOOCV pipeline with 5,000 label permutations; empirical p-values were FWE-corrected using a max-statistic approach. All analyses were conducted in R. We chose univariate rather than multivariable/machine-learning models owing to (i) overfitting risk in a modest sample despite regularization, (ii) better interpretability with clear effect sizes and clinically interpretable thresholds, and (iii) our aim of hypothesis testing rather than optimizing prediction. Future studies with larger cohorts will be able to compare multivariable and machine-learning approaches.

We also compared three multivariable models built from all available features: a clinical-only model using all clinical variables, an FC-only model using all FC connections, and a combined model including both clinical and FC features. The clinical-only model used standard logistic regression. The FC-only and combined models used elastic-net regularization with an inner cross-validation to select the penalty; in the combined model, clinical terms were left unpenalized while FC terms were penalized. Discrimination was assessed by area under the curves (AUCs) and 95% confidence intervals (CIs), and models were compared using pairwise DeLong tests, and incremental value by the category-free net reclassification improvement (NRI) and the integrated discrimination improvement (IDI).

Statistical analyses

Chi-square tests, ANOVA tests and independent samples t-tests were used to analyze the demographic and clinical characteristics of the participants using the SPSS software (version 27.0) with a significant threshold of $p < 0.05$ (two-tailed).

Imaging data analyses were conducted using CONN Toolbox. FC values of significant clusters were extracted for each group. Post hoc pairwise comparisons were performed in SPSS with Bonferroni correction, applying a significance threshold of $p < 0.016$ (i.e., $0.05/3$).

Pearson correlation analyses were conducted to assess the associations between differential FC values and clinical features, including baseline Y-BOCS scores and Y-BOCS reduction scores (Δ Y-BOCS), across and within nOCD and rOCD groups. Statistical significance was defined as $p < 0.05$ (two-tailed). We further evaluated whether baseline depressive or anxiety symptom severity would moderate the association between FC and sertraline response. For each FC, logistic regression models were fitted including the FC main effect, baseline BDI or BAI scores, and an interaction term ($FC \times BDI$ or $FC \times BAI$), with p-values adjusted for multiple comparisons using false discovery rate (FDR) corrections.

Results

Demographic and clinical characteristics

Demographic and clinical characteristics are summarized in Table 1 and Supplementary Table 2. Regarding comorbidity, 5 of 54 OCD patients (1 in the nOCD group and 4 in the rOCD group) self-reported a prior depressive disorder, and 2 patients in the rOCD group reported a prior anxiety disorder. No significant differences were observed among nOCD, rOCD and HC groups in terms of sex, age or years of education, or between the nOCD and rOCD groups specifically. Additionally, the nOCD and rOCD groups did not differ in illness duration, baseline Y-BOCS, BDI or BAI scores.

Table 1. Demographic and clinical characteristics of the study participants.

Variables (mean $\pm$ SD)	nOCD(n=21)	rOCD(n=33)	HC(n=39)	$t/F/\chi^2$	p -Value
Sex (male/female)	15/6	19/14	19/20	2.879	0.237 ^a
Age (years)	2	2	3	1.778	0.175 ^b
Education (years)	1	1	1	0.082	0.922 ^b
Illness Duration (years)	1	8	NA	0.916	0.364 ^c
Y-BOCS score before Medication	2	2	NA	1.545	0.128 ^c
Y-BOCS score after Medication	2	1	NA	-	-
BDI	1	2	NA	1.397	0.168 ^c
BAI	1	1	NA	1.888	0.065 ^c

OCD, obsessive-compulsive disorder; nOCD, non-response OCD patients; rOCD, response OCD patients; HC, healthy control; Y-BOCS, the Yale-Brown Obsessive Compulsive Scale; BDI, Beck Depression Inventory; BAI, Beck Anxiety Inventory; NA, not applicable.

^a p -value from the chi-square test; ^b p -value from one-way analysis of variance tests; ^c p -value from independent samples t-tests (nOCD vs. rOCD).

Seed-based functional connectivity analyses

We first compared FC between the OCD group (nOCD and rOCD) and HC (Supplementary Table 3), and then we conducted a one-way ANOVA of seed-based FC among all three groups (voxel-wise FWE-corrected, $p < 0.05$). followed by Bonferroni-corrected post hoc tests ($p < 0.05$) for pairwise comparisons. Table 2 summarizes post hoc differences in FC between nOCD and rOCD groups.

ANOVA results are shown in Figure 1 and Supplementary Table 4. To avoid repetition, all abbreviations for seed regions and their corresponding brain areas are

listed in Supplementary Table 5. In the left hemisphere, the altered FC included Thal_l-FG_r, Thal_l-PoG_l, SomMotA_1_l-VIIb_l, SomMotA_1_l-VI_l, SomMotA_1_l-CC, SomMotA_2_l-STN, SomMotA_2_l-VI_l, SomMotA_2_l-Vermis, SomMotA_2_l-Pu, SomMotA_4_l-VPL_r and SomMotA_4_l-Caud_r. In the right hemisphere, the altered FC included Thal_r-Crus1_r, Thal_r-PoG_l, SomMotB_4_r-MTG_r and SomMotB_4_r-MOG_l. No significant difference was found when the right putamen was used as the seed.

Table 2. Brain regions demonstrating different FC values between nOCD and rOCD groups.

Seed region	Brain region	MNI peak			Cluster size	F-value (peak)	Post hoc analyses
		X	Y	Z			
Thalamus_l	Fusiform Gyrus_r	26	-80	-04	129	8.511	nOCD < rOCD, rOCD > HC
	Postcentral Gyrus_r	-44	-28	68	122	9.600	nOCD < rOCD, rOCD > HC
Putamen_l	Extra-nuclear White Matter_r	10	-34	04	86	9.153	nOCD > rOCD, rOCD < HC
SomMotA_1_l	Cerebellar Lobule VIIb_l	-06	-72	-36	439	10.192	nOCD < rOCD, rOCD > HC
	Cerebellar Lobule VI_l	-26	-58	-28	141	8.421	nOCD < rOCD, nOCD < HC
	Corpus Callosum_l	-22	-48	06	122	10.736	nOCD > rOCD, rOCD < HC
SomMotA_2_l	Subthalamic Nucleus_r	10	-14	-08	184	9.536	nOCD < rOCD, rOCD > HC
	Cerebellar Lobule VI_l	-26	-58	-28	167	10.044	nOCD < rOCD, nOCD < HC
	Cerebellar Vermis VII_r	06	-64	-26	85	8.103	nOCD < rOCD
	Pulvinar_l	-22	-28	12	83	8.334	nOCD < rOCD, rOCD > HC
SomMotA_4_l	Ventral Posterolateral Nucleus_r	16	-18	04	129	8.511	nOCD < rOCD, rOCD > HC
	Caudate_r	20	-02	18	87	9.276	nOCD < rOCD, rOCD > HC
Thalamus_r	Cerebellar Crus I_r	40	-68	-38	264	9.144	nOCD > rOCD, nOCD > HC
	Postcentral Gyrus_l	-44	-28	68	116	8.486	nOCD < rOCD
SomMotB_4_r	Middle Temporal Gyrus_r	46	-70	-02	140	7.582	nOCD < rOCD
	Middle Occipital Gyrus_l	-38	-74	00	108	9.476	nOCD < rOCD, rOCD > HC

MNI Peak, Coordinates of primary peak locations in the Montreal Neurological Institute space; l, left; r, right; SomMot, somatomotor network; nOCD, non-response OCD patients; rOCD, response OCD patients; HC, healthy control.

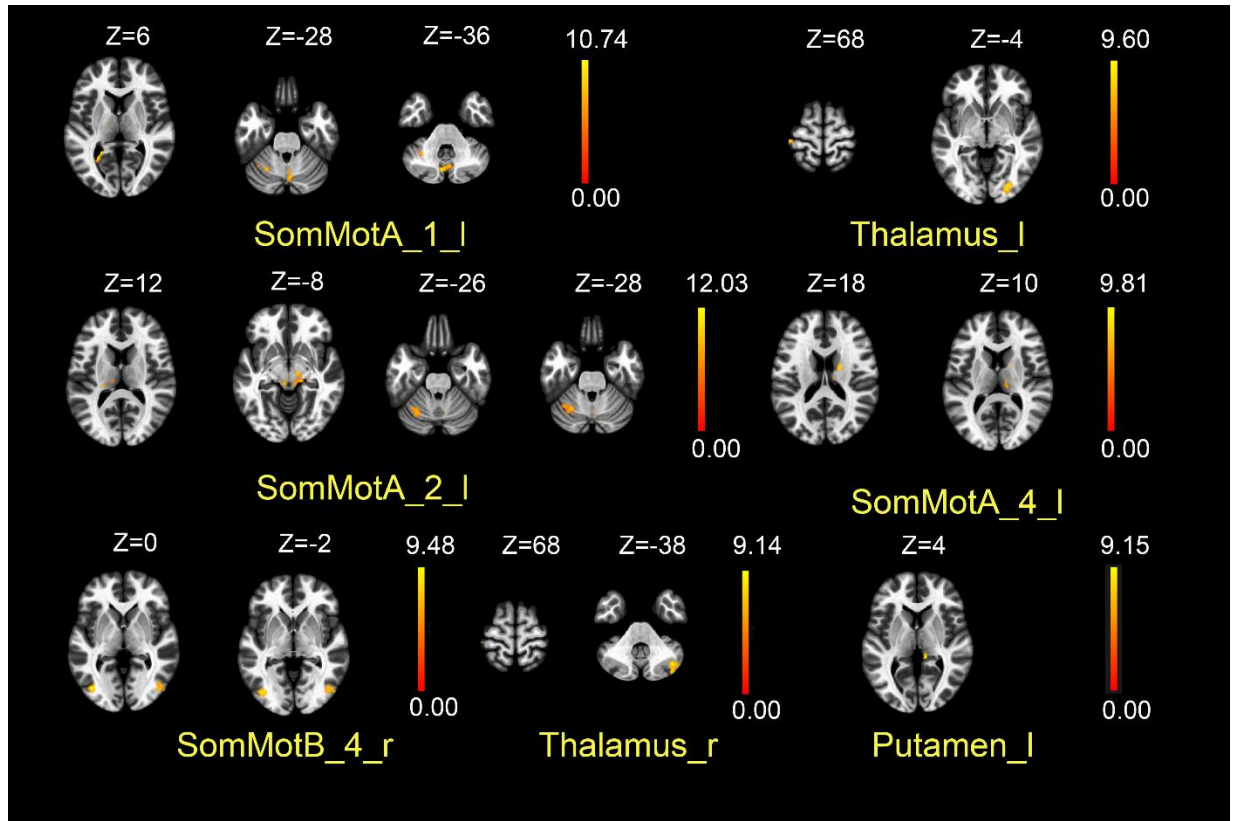


Figure 1. Brain regions with significant FC differences among the nOCD, rOCD and HC groups based on ANOVA analysis. The color bars indicate the F -value. SomMot, somatomotor network; l, left; r, right.

Post hoc analyses

Bonferroni-corrected post hoc comparisons are summarized in Table 2 (complete results in Supplementary Table 6). Across all analyses, we found a consistent pattern of altered FC within the sensorimotor circuit (Figure 2A). Compared to HC, the rOCD group showed significantly **higher** FC including Thal_l-PoG_l, SomMotA_2_l-STN, SomMotA_4_l-VPL_r and SomMotA_4_l-Caud_r. Although the difference did not reach statistical significance, the connectivity between Thal_r and PoG_l **was also higher** in the rOCD group relative to HC. In contrast, all of these FC values were **lower** in the nOCD group than in HCs, although the differences did not reach statistical significance.

Another finding was the presence of several altered FC between the sensorimotor circuit and cerebellum (Figure 2B). The nOCD group showed a significantly **higher** connectivity between Thal_r and Crus1_r, and significantly **lower** FC values including SomMotA_1_l-VI_l and SomMotA_2_l-VI_l, relative to the HC group. In the rOCD group, the connectivity between SomMotA_1_l and VIIb_l was significantly **higher** compared to HC group.

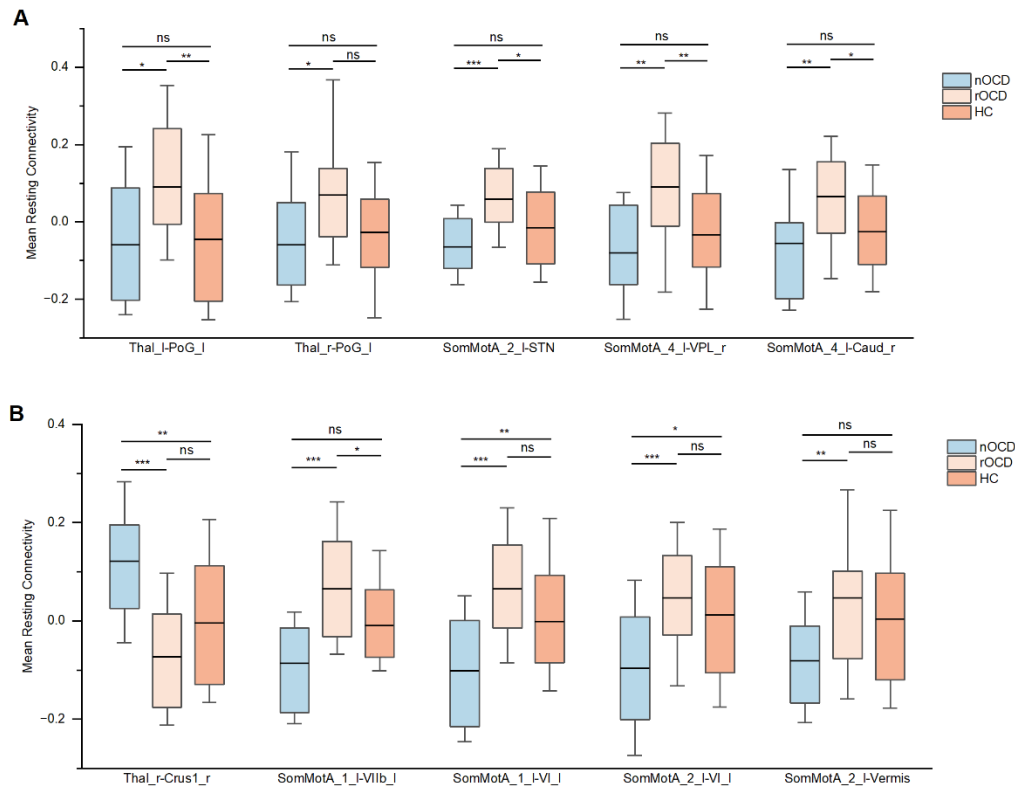


Figure 2. Bar plots for partial results of post hoc analyses of FC values between each pair group (nOCD group vs. rOCD group, nOCD group vs. HC, rOCD group vs. HC). The y-axis shows the values extracted from the significant FC, the central line represents the mean value, the box spans from the 25th to the 75th percentile, and the whiskers extend from the 10th to the 90th percentile. (A) Differences in FC within the sensorimotor circuit. (B) Differences in FC between the sensorimotor circuit and cerebellum. Significance coding: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns $p > 0.05$. Thal, thalamus; PoG, postcentral gyrus; SomMot, somatomotor network; STN, subthalamic nucleus; VPL, ventral posterolateral nucleus; Caud, caudate; Crus1, cerebellar Crus I; VIIb, cerebellar lobule VIIb; VI, cerebellar lobule VI; Vermis, cerebellar vermis VII; nOCD, non-response OCD patients; rOCD, response OCD patients; HC, healthy control; l, left; r, right.

Correlation analysis between functional connectivity and clinical features

To explore the relationship between altered FC and clinical features in OCD patients, we conducted Pearson correlation analyses both across and within the nOCD and rOCD groups. No significant correlations were found between baseline Y-BOCS scores and any altered FC either across the two groups or within the rOCD group. However, in the nOCD group, baseline Y-BOCS scores were positively correlated with FC values between SomMotA_1_l and VI_l (Supplementary Figure 2). Several

correlations were identified between ΔY -BOCS scores and differential FC (Supplementary Figure 3), with most of the associations being positive.

We next examined whether baseline depressive or anxiety symptom severity moderated the association between FC and sertraline response. Results showed that none of the interaction terms ($FC \times BDI$ or $FC \times BAI$) reached significance after FDR correction (Supplementary Tables 7 and 8). We found no evidence that baseline depressive or anxiety symptom severity substantially modified the relationship between FC and sertraline response in this sample.

Exploratory prediction of sertraline response

To test whether the FC that differed between the rOCD and nOCD groups could predict sertraline response, we first conducted univariate logistic regressions for each FC. In the apparent analyses, no clinical variables showed significant predictive value after FDR correction, whereas some altered FC exhibited large apparent effects, with R^2 ranging from 0.144 to 0.476 (Figure 3A and Supplementary Table 9).

We then applied LOOCV to mitigate optimism, with covariate adjustment performed within each fold. Notably, a specific FC of the sensorimotor circuit, Thal_r-Crus1_r, was the only predictor to achieve statistical significance ($p = 0.010$) after FWE-corrected permutation tests. This model demonstrated strong discriminatory ability ($AUC = 0.854$, Figure 3B and Supplementary Figure 4). All other FC-based predictors showed better performances than clinical features, though none of them reached statistical significance (Supplementary Table 10).

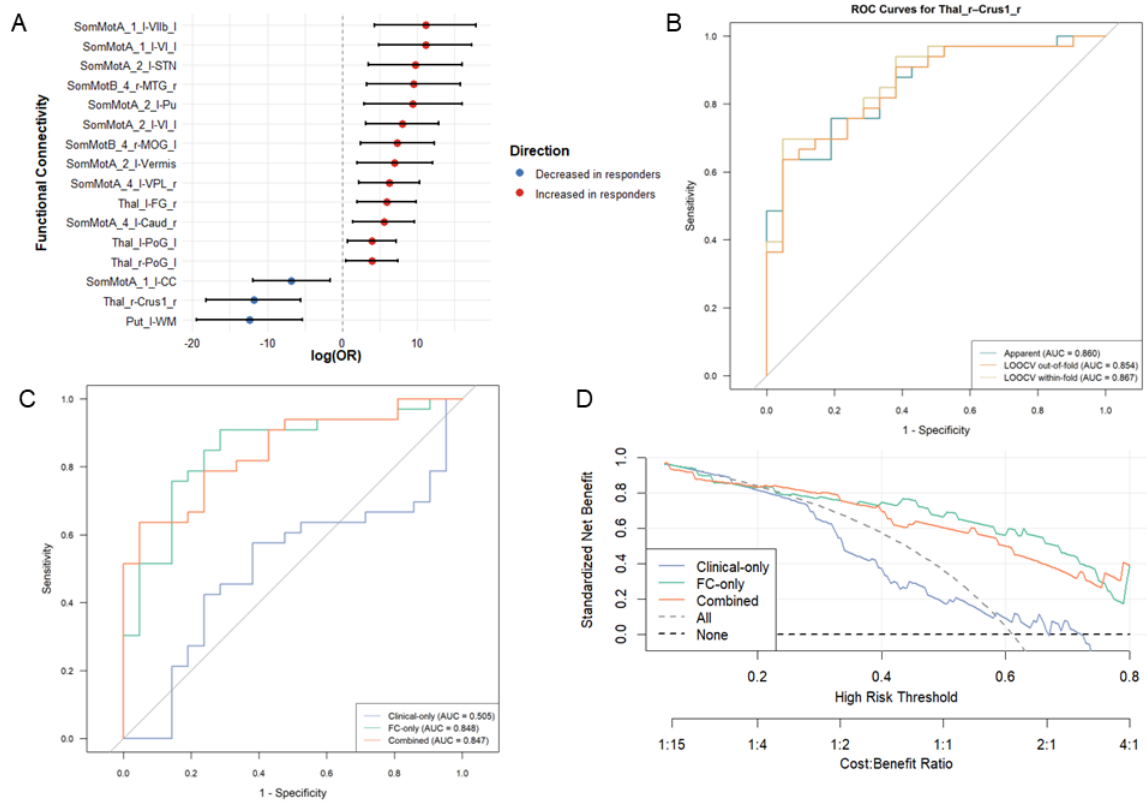


Figure 3. Exploratory prediction of sertraline response in OCD patients. (A) Forest plot showing ORs and 95% CI for altered FC between rOCD and nOCD groups identified by binary logistic regression analyses. FC associated with treatment response is shown in red, and those associated with non-response are shown in blue. (B) The apparent (optimistic), LOOCV out-of-fold and LOOCV within-fold ROC curves for Thal_r-Crus1_r. (C) ROC curves for clinical-only, FC-only and combined models. (D) Decision-curve analysis. Standardized net benefit is plotted against the threshold probability from 0.05 to 0.80, with treat-all (gray dashed) and treat-none (black dashed) reference strategies. OR, the odds ratio; ROC, receiver operating characteristic; AUC, area under the curve; Thal, thalamus; Crus1, cerebellar Crus I; PoG, postcentral gyrus; Put, putamen; WM, white matter; FG, fusiform gyrus; SomMot, somatomotor network; VIIb, cerebellar lobule VIIb; VI, cerebellar lobule VI; CC, corpus callosum; STN, subthalamic nucleus; Vermis, cerebellar vermis VII; Pu, pulvinar; VPL, ventral posterolateral nucleus; Caud, caudate; MTG, middle temporal gyrus; MOG, middle occipital gyrus; l, left; r, right.

Sensitivity to response threshold

We repeated the cross-validated analyses under different cut-offs of Y-BOCS score reduction (25%, 30%, 50%, Supplementary Table 11). At a 30% cut-off, overall discriminability was slightly attenuated relative to the primary 35% analysis, but Thal_r-Crus1_r and SomMotA_1_l-VIIb_l still showed above-chance out-of-sample

performance. In contrast, at 25% (more liberal) and 50% (more stringent) cut-offs, most FC yielded LOOCV AUCs near chance and were not significant after FDR correction. This likely reflects more misclassification at 25% and reduced power at 50% due to fewer responders.

Comparison to clinical predictors

To evaluate whether FC outperforms clinical predictors, we fitted three models: (A) clinical-only, (B) FC-only, and (C) combined (clinical + FC). Under LOOCV, the clinical-only model performed at chance (AUC = 0.505, 95% CI 0.343–0.668), whereas the FC-only (AUC = 0.848, 95% CI 0.740–0.957) and combined (AUC = 0.847, 95% CI 0.746–0.948) models showed markedly higher discrimination (Figure 3C). DeLong tests indicated that both FC-only and combined models had significantly higher AUCs than the clinical-only model, with no significant difference between them (Supplementary Table 12). Category-free NRI and IDI likewise showed a substantial improvement of the combined model over the clinical-only model but only minimal, non-significant gain over the FC-only model (Supplementary Table 12).

Decision-curve analysis showed that across threshold probabilities of 0.20–0.80, the FC-only model achieved the highest (or equal) net benefit. The combined model provided a small additional net benefit over the FC-only model only within a narrow range around a threshold probability of 0.30, whereas the clinical-only model yielded the lowest net benefit and became negative at higher thresholds (Figure 3D). Together, in this cohort, FC features provided the main clinical gain, and the combined model offered only modest incremental value within a narrow threshold range.

Discussion

Reliable predictors for SSRI response could help reduce disability and socioeconomic burden of OCD patients. In this exploratory study focusing on the sensorimotor circuit, we observed distinct baseline FC patterns across rOCD, nOCD, and HC groups. Altered FC mainly involved connections within the sensorimotor circuit, as well as between the circuit and the cerebellum. In predictive analyses, these sensorimotor FC features discriminated sertraline responders from non-responders better than baseline clinical characteristics, suggesting their potential as candidate neural markers of sertraline response. However, this was a single-site exploratory study with sample falls below recent recommendations for robust brain–behavior prediction studies ($n \geq 100$)[39], and no external validation was available. Accordingly, these findings are best viewed as hypothesis-generating and will require replication in larger, independent cohorts before any clinical implementation.

Distinct FC patterns within the sensorimotor circuit

Within the sensorimotor circuit, FC differences between rOCD and nOCD groups displayed a consistent pattern. Most FC values in the rOCD group were significantly **higher** than in HCs, whereas the nOCD group tended to show **lower** FC than HCs, though not statistically significant. Altered connectivity was observed between the bilateral thalamus and left PoG, as well as between the left sensorimotor cortices and subcortical structures in this circuit, such as STN, VPL and caudate, which play roles in the sensorimotor process[40-42]. The differential patterns of FC observed between responders and non-responders may partially explain the heterogeneity in previous studies investigating FC within the sensorimotor circuit in OCD patients. Several rs-fMRI studies, including the research from the ENIGMA consortium, have observed **lower** FC within the sensorimotor regions[13,43-45]. Conversely, **higher** FC values within sensorimotor circuits[46,47], such as elevated FC values between the left STN and right premotor cortex[48], have also been reported. Such discrepancies may reflect differences in analytic strategies or medication status, but our results also suggest that baseline FC abnormalities within the sensorimotor circuit may be heterogeneous across patients.

We hypothesize that these baseline FC differences may reflect variation in illness subtypes or in the level of neural reserve within the sensorimotor circuit. Higher FC in the rOCD group suggest relatively preserved or compensatory neural resources within this circuit, whereas lower FC in the nOCD group may signal a more disintegrated or compromised state. This interpretation is consistent with findings in unaffected siblings of OCD patients, who showed hyperactivation of the presupplementary motor area (pre-SMA) during response inhibition—even exceeding that of OCD probands—while maintaining preserved behavioral performance, a pattern interpreted as compensatory recruitment of control circuitry in the context of familial risk[49]. In line with this, higher sensorimotor FC in rOCD group may reflect a more protective or compensatory pattern of coupling. In healthy volunteers and patients with depression, SSRI treatment has been linked to changes in activity and connectivity within large-scale brain networks including sensorimotor circuit[23,50-52]. On this basis, we further speculate that better functional preservation may facilitate SSRI-related neuroplastic processes and thereby enhance their therapeutic effects. This proposed mechanism remains hypothetical and validation through longitudinal and multimodal investigations will be essential to clarify the neurobiological basis of treatment heterogeneity in OCD.

Cerebellar-sensorimotor connectivity in OCD

Another notable finding was the differential FC between the cerebellum and the sensorimotor circuit, highlighting a potentially underappreciated cerebellum contribution to OCD. Altered connectivity was observed between the right thalamus and cerebellum Crus I, as well as between left sensorimotor cortices and cerebellar lobules VIIb/VI and vermis VII. Beyond its established role in sensorimotor control[53,54], but accumulating evidence suggests that it also contributes to higher-level brain functions, including cognition, emotion regulation, and language processing[55,56]. Accordingly, it has emerged as a novel and increasingly important target of investigation in OCD. Prior studies found that cerebellar lesions can trigger the onset of OCD symptoms[57]. In addition, neuroimaging studies have revealed cerebellar structural and functional abnormalities in OCD patients, including increased gray matter volume[58], reduced fractional anisotropy[59], and altered regional spontaneous brain activity[60]. However, current research investigating the relationship between SSRI treatment efficacy and cerebellar-cerebral networks remains limited. Our findings suggest that connectivity between the cerebellum and the sensorimotor circuit may play a role in the neural regulation underlying treatment response in OCD. Given the capacity for forward modulation of cerebellum[61,62], we hypothesize that its integrative and regulatory influence over sensorimotor and other brain networks may be important for SSRI efficacy, while it warrants further empirical validation.

Predictive utility and clinical translation

Identifying reliable predictors of SSRI treatment response in OCD remains a major clinical challenge. In this study, each FC that differed between the rOCD and nOCD groups achieved a LOOCV AUC above chance when used to distinguish sertraline responders from non-responders. Specifically, one connection of the sensorimotor circuit, Thal_r–CrusI_r, remained predictive even after stringent correction. Moreover, in multivariable analyses, the FC-only model outperformed the clinical-only model in this cohort, and achieved similar performance to the combined (clinical + FC) model, suggesting that incorporating FC features with standard clinical characteristics may enhance individualized prediction and risk stratification. Taken together, these findings suggest FC of the sensorimotor circuit may serve as a promising candidate biomarker of sertraline response in OCD. Given the modest sample size and absence of independent validation, the present results should still be regarded as exploratory, providing a basis for future, larger-scale studies.

Limitation

Several limitations should be noted. First, this single-site, single-scanner, baseline-only rs-fMRI study in a Chinese sample was cross-sectional, which precludes causal inference and limits generalizability to other populations, scanners and health-care settings. Baseline FC differences may reflect different OCD subtypes, variations in neural reserve, pre-existing severity differences not captured by the Y-BOCS, or other unmeasured confounds rather than pathways of SSRI action. Second, we lacked specific clinical or cognitive measures of sensorimotor-related phenomena, and no post-treatment imaging was acquired, preventing us from directly linking FC abnormalities to sensorimotor symptoms or to longitudinal changes under sertraline. Third, only sertraline was examined, final dose data were missing for some patients, and adherence were assessed mainly by self-report, so inadequate exposure cannot be excluded and the present data cannot determine whether these FC features mark SSRI-specific or more general treatment response. Fourth, the relatively small and unbalanced sample, absence of an external validation cohort mean that prediction metrics are likely optimistic and residual false positives cannot be ruled out, despite cross-validation and permutation testing. Together, these limitations indicate that our findings should be interpreted as exploratory and hypothesis-generating, and require replication and extension in larger, multi-site samples with longitudinal imaging and more detailed clinical cohorts. Based on the cross-validated AUC of approximately 0.75, an external validation study would require about 107 participants to achieve 80% power and 141 to achieve 90% power at $\alpha = 0.05$; Assuming a 10% lower true AUC, the required sample sizes increase to approximately 241 and 323, respectively.

Conclusion

In conclusion, baseline FC of the sensorimotor circuit differed between sertraline responders and non-responders among drug-naïve OCD patients and showed potential to discriminate between these groups. These findings support the sensorimotor circuit as a candidate neural marker of sertraline treatment outcome in OCD, but should be regarded as exploratory and require validation in larger, independent cohorts before clinical implementation.

CRedit Authorship Contribution Statement

Tingyue Bai: Writing – original draft, Writing – review & editing, Formal analysis, Investigation, Data curation; **Hanyang Ruan:** Writing – review & editing, Conceptualization, Methodology, Validation; **Yang Wang:** Writing – review & editing, Conceptualization, Methodology; **Gui Zhang:** Conceptualization; **Jian Gao:**

Investigation; **Zifeng Zheng:** Investigation; **Di Li:** Investigation; **Qing Zhao:** Investigation; **Huiqin Han:** Resources; **Qing Fan:** Resources; **Dongdong Shi:** Resources; **Zhen Wang:** Supervision, Resources, Funding acquisition, Conceptualization.

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Data availability statement

Individual-level clinical and imaging data derive from an ongoing longitudinal study, and therefore they cannot be publicly shared until the cohort's primary results are published. Derived measures and analysis code are available from the corresponding author upon reasonable request.

Competing Interests

The authors declare no competing interests.

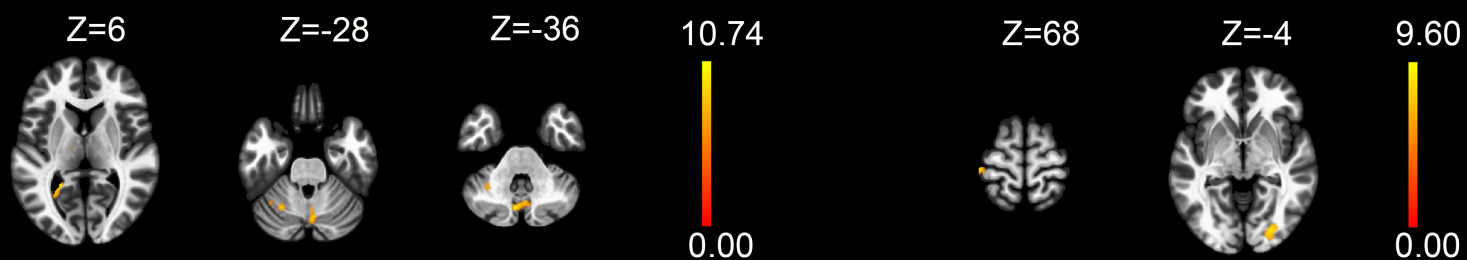
- 1 Goodman WK, Storch EA, Sheth SA. Harmonizing the Neurobiology and Treatment of Obsessive-Compulsive Disorder. *The American journal of psychiatry*. 2021;178(1):17-29.
- 2 Ruscio AM, Stein DJ, Chiu WT, Kessler RC. The epidemiology of obsessive-compulsive disorder in the National Comorbidity Survey Replication. *Molecular psychiatry*. 2010;15(1):53-63.
- 3 Huang Y, Wang Y, Wang H, Liu Z, Yu X, Yan J, et al. Prevalence of mental disorders in China: a cross-sectional epidemiological study. *Lancet Psychiatry*. 2019;6(3):211-24.
- 4 Adam Y, Meinlschmidt G, Gloster AT, Lieb R. Obsessive-compulsive disorder in the community: 12-month prevalence, comorbidity and impairment. *Soc Psychiatry Psychiatr Epidemiol*. 2012;47(3):339-49.
- 5 Yang W, Tang Z, Wang X, Ma X, Cheng Y, Wang B, et al. The cost of obsessive-compulsive disorder (OCD) in China: a multi-center cross-sectional survey based on hospitals. *Gen Psychiatr*. 2021;34(6):e100632.
- 6 Romanelli RJ, Wu FM, Gamba R, Mojtabai R, Segal JB. Behavioral therapy and serotonin reuptake inhibitor pharmacotherapy in the treatment of obsessive-compulsive disorder: a systematic review and meta-analysis of head-to-head randomized controlled trials. *Depress Anxiety*. 2014;31(8):641-52.
- 7 Simpson HB, Huppert JD, Petkova E, Foa EB, Liebowitz MR. Response versus remission in obsessive-compulsive disorder. *J Clin Psychiatry*. 2006;67(2):269-76.
- 8 Hazari N, Narayanaswamy JC, Arumugham SS. Predictors of response to serotonin reuptake inhibitors in obsessive-compulsive disorder. *Expert Rev Neurother*. 2016;16(10):1175-91.
- 9 Stein DJ, Costa DLC, Lochner C, Miguel EC, Reddy YCJ, Shavitt RG, et al. Obsessive-compulsive disorder. *Nat Rev Dis Primers*. 2019;5(1):52.
- 10 Nakao T, Okada K, Kanba S. Neurobiological model of obsessive-compulsive disorder: evidence from recent neuropsychological and neuroimaging findings. *Psychiatry Clin Neurosci*. 2014;68(8):587-605.
- 11 van den Heuvel OA, van Wingen G, Soriano-Mas C, Alonso P, Chamberlain SR, Nakamae T, et al. Brain circuitry of compulsivity. *Eur Neuropsychopharmacol*. 2016;26(5):810-27.
- 12 Shephard E, Stern ER, van den Heuvel OA, Costa DLC, Batistuzzo MC, Godoy PBG, et al. Toward a neurocircuit-based taxonomy to guide treatment of obsessive-compulsive disorder. *Molecular psychiatry*. 2021;26(9):4583-604.
- 13 Bruin WB, Abe Y, Alonso P, Anticevic A, Backhausen LL, Balachander S, et al. The functional connectome in obsessive-compulsive disorder: resting-state mega-analysis and machine learning classification for the ENIGMA-OCD consortium. *Molecular psychiatry*. 2023;28(10):4307-19.
- 14 Abramovitch A, Abramowitz JS, Mittelman A. The neuropsychology of adult obsessive-compulsive disorder: a meta-analysis. *Clinical psychology review*. 2013;33(8):1163-71.
- 15 Lipszyc J, Schachar R. Inhibitory control and psychopathology: a meta-analysis of

- studies using the stop signal task. *J Int Neuropsychol Soc.* 2010;16(6):1064-76.
- 16 Zandbelt BB, Bloemendaal M, Hoogendam JM, Kahn RS, Vink M. Transcranial magnetic stimulation and functional MRI reveal cortical and subcortical interactions during stop-signal response inhibition. *J Cogn Neurosci.* 2013;25(2):157-74.
- 17 Rodriguez-Manrique D, Ruan H, Winkelmann C, Haun J, Gigl S, Berberich G, et al. Investigating the effects of brain stimulation on the neural substrates of inhibition in patients with OCD: A simultaneous tDCS - fMRI study. *Translational psychiatry.* 2025;15(1):173.
- 18 Bär KJ, de la Cruz F, Schumann A, Koehler S, Sauer H, Critchley H, et al. Functional connectivity and network analysis of midbrain and brainstem nuclei. *Neuroimage.* 2016;134:53-63.
- 19 Beliveau V, Svarer C, Frokjaer VG, Knudsen GM, Greve DN, Fisher PM. Functional connectivity of the dorsal and median raphe nuclei at rest. *Neuroimage.* 2015;116:187-95.
- 20 Shanahan NA, Holick Pierz KA, Masten VL, Waeber C, Ansorge M, Gingrich JA, et al. Chronic reductions in serotonin transporter function prevent 5-HT1B-induced behavioral effects in mice. *Biological psychiatry.* 2009;65(5):401-8.
- 21 Woehrle NS, Klenotich SJ, Jamnia N, Ho EV, Dulawa SC. Effects of chronic fluoxetine treatment on serotonin 1B receptor-induced deficits in delayed alternation. *Psychopharmacology (Berl).* 2013;227(3):545-51.
- 22 Pittenger C, Adams TG, Jr., Gallezot JD, Crowley MJ, Nabulsi N, James R, et al. OCD is associated with an altered association between sensorimotor gating and cortical and subcortical 5-HT1b receptor binding. *Journal of affective disorders.* 2016;196:87-96.
- 23 Melo L, Beaupain MC, Ghanavati E, Kuo MF, Nitsche MA. Neurochemical mechanisms underlying serotonergic modulation of neuroplasticity in humans. *Brain Stimul.* 2024;17(2):421-30.
- 24 Kim M, Kwak S, Yoon YB, Kwak YB, Kim T, Cho KIK, et al. Functional connectivity of the raphe nucleus as a predictor of the response to selective serotonin reuptake inhibitors in obsessive-compulsive disorder. *Neuropsychopharmacology.* 2019;44(12):2073-81.
- 25 Yan H, Shan X, Li H, Liu F, Xie G, Li P, et al. Cerebellar functional connectivity and its associated genes: A longitudinal study in drug-naïve patients with obsessive-compulsive disorder. *Journal of psychiatric research.* 2024;177:378-91.
- 26 Yan H, Zhang Y, Shan X, Li H, Liu F, Xie G, et al. Altered interhemispheric functional connectivity in patients with obsessive-compulsive disorder and its potential in therapeutic response prediction. *J Neurosci Res.* 2024;102(1).
- 27 Wang H, Teng C, Zhang D, Zhang W, Zhang H, Wu X, et al. Abnormal intrinsic brain activity of the sensory-motor area as a predictor of the response to selective serotonin reuptake inhibitors in treatment-naïve obsessive-compulsive disorder. *Journal of affective disorders.* 2025:119457.
- 28 Association AP. Diagnostic and statistical manual of mental disorders: DSM-5. American psychiatric association; 2013.

- 29 Goodman WK, Price LH, Rasmussen SA, Mazure C, Fleischmann RL, Hill CL, et al. The Yale-Brown Obsessive Compulsive Scale. I. Development, use, and reliability. *Arch Gen Psychiatry*. 1989;46(11):1006-11.
- 30 Sheehan DV, Lecrubier Y, Sheehan KH, Amorim P, Janavs J, Weiller E, et al. The Mini-International Neuropsychiatric Interview (M.I.N.I.): the development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. *J Clin Psychiatry*. 1998;59 Suppl 20:22-33;quiz 34-57.
- 31 Wang P, Gu W, Gao J, Wang C, Fang J, Hu M, et al. Protocol for a Pragmatic Trial of Pharmacotherapy Options Following Unsatisfactory Initial Treatment in OCD (PROCEED). *Frontiers in psychiatry*. 2022;13:822976.
- 32 Zheng YP, Wei LA, Goa LG, Zhang GC, Wong CG. Applicability of the Chinese Beck Depression Inventory. *Comprehensive psychiatry*. 1988;29(5):484-9.
- 33 Che H-H, Lu M-L, Chen H-C, Chang S-W, Lee Y-J. Validation of the Chinese Version of the Beck Anxiety Inventory. *Formosan Journal of Medicine*. 2006;10:447-54.
- 34 Farris SG, McLean CP, Van Meter PE, Simpson HB, Foa EB. Treatment response, symptom remission, and wellness in obsessive-compulsive disorder. *J Clin Psychiatry*. 2013;74(7):685-90.
- 35 Nieto-Castanon A, Whitfield-Gabrieli S.
- 36 Penny WD, Friston KJ, Ashburner JT, Kiebel SJ, Nichols TE. Statistical parametric mapping: the analysis of functional brain images. Elsevier; 2011.
- 37 Kong R, Yang Q, Gordon E, Xue A, Yan X, Orban C, et al. Individual-Specific Areal-Level Parcellations Improve Functional Connectivity Prediction of Behavior. *Cerebral Cortex*. 2021;31(10):4477-500.
- 38 Desikan RS, Ségonne F, Fischl B, Quinn BT, Dickerson BC, Blacker D, et al. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *Neuroimage*. 2006;31(3):968-80.
- 39 Marek S, Laumann TO. Replicability and generalizability in population psychiatric neuroimaging. *Neuropsychopharmacology*. 2024;50(1):52-57.
- 40 van Wijk BCM, Alkemade A, Forstmann BU. Functional segregation and integration within the human subthalamic nucleus from a micro- and meso-level perspective. *Cortex*. 2020;131:103-13.
- 41 Grahn JA, Parkinson JA, Owen AM. The cognitive functions of the caudate nucleus. *Prog Neurobiol*. 2008;86(3):141-55.
- 42 Li Z, Luo J, Li C, Zhu H. Upregulation of Nav1.6 expression in the ventral posterolateral nucleus of thalamus contributes to hyperalgesia in a model of Parkinson's disease. *Exp Neurol*. 2025;383:115032.
- 43 Lv D, Ou Y, Chen Y, Ding Z, Ma J, Zhan C, et al. Anatomical distance affects functional connectivity at rest in medicine-free obsessive-compulsive disorder. *BMC psychiatry*. 2022;22(1):462.
- 44 Xu Y, Zhang Y. Abnormal voxel-mirrored homotopic connectivity in first-episode, drug-naïve patients with obsessive-compulsive disorder. *Eur J Neurosci*. 2023;58(6):3531-39.
- 45 Stern ER, Eng GK, De Nadai AS, Iosifescu DV, Tobe RH, Collins KA. Imbalance

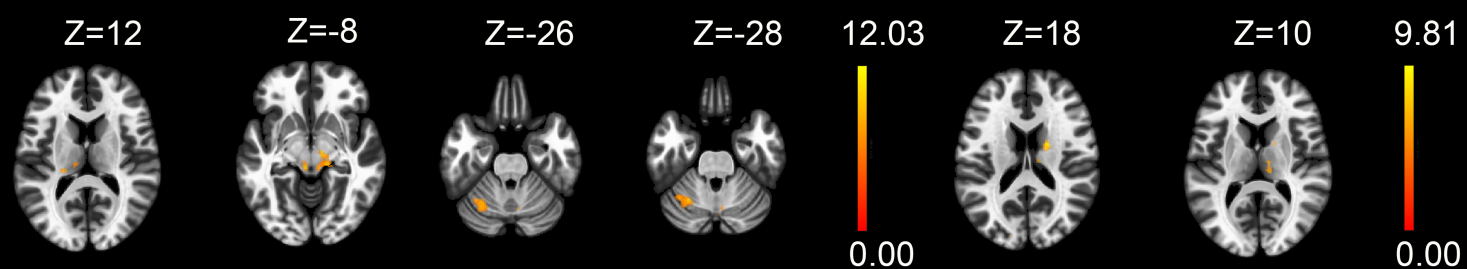
- between default mode and sensorimotor connectivity is associated with perseverative thinking in obsessive-compulsive disorder. *Translational psychiatry*. 2022;12(1):19.
- 46 Chen Y, Meng Z, Zhang Z, Zhu Y, Gao R, Cao X, et al. The right thalamic glutamate level correlates with functional connectivity with right dorsal anterior cingulate cortex/middle occipital gyrus in unmedicated obsessive-compulsive disorder: A combined fMRI and (1)H-MRS study. *Aust N Z J Psychiatry*. 2019;53(3):207-18.
- 47 Tikoo S, Cardona F, Tommasin S, Giannì C, Conte G, Upadhyay N, et al. Resting-state functional connectivity in drug-naïve pediatric patients with Tourette syndrome and obsessive-compulsive disorder. *Journal of psychiatric research*. 2020;129:129-40.
- 48 Cano M, Alonso P, Martínez-Zalacain I, Subirà M, Real E, Segalàs C, et al. Altered functional connectivity of the subthalamus and the bed nucleus of the stria terminalis in obsessive-compulsive disorder. *Psychological medicine*. 2018;48(6):919-28.
- 49 de Wit SJ, de Vries FE, van der Werf YD, Cath DC, Heslenfeld DJ, Veltman EM, et al. Presupplementary motor area hyperactivity during response inhibition: a candidate endophenotype of obsessive-compulsive disorder. *The American journal of psychiatry*. 2012;169(10):1100-8.
- 50 Klaassens BL, van Gorsel HC, Khalili-Mahani N, van der Grond J, Wyman BT, Whitcher B, et al. Single-dose serotonergic stimulation shows widespread effects on functional brain connectivity. *Neuroimage*. 2015;122:440-50.
- 51 Chau DT, Fogelman P, Nordanskog P, Drevets WC, Hamilton JP. Distinct Neural-Functional Effects of Treatments With Selective Serotonin Reuptake Inhibitors, Electroconvulsive Therapy, and Transcranial Magnetic Stimulation and Their Relations to Regional Brain Function in Major Depression: A Meta-analysis. *Biol Psychiatry Cogn Neurosci Neuroimaging*. 2017;2(4):318-26.
- 52 Lesch KP, Waider J. Serotonin in the modulation of neural plasticity and networks: implications for neurodevelopmental disorders. *Neuron*. 2012;76(1):175-91.
- 53 De Zeeuw CI, Lisberger SG, Raymond JL. Diversity and dynamism in the cerebellum. *Nat Neurosci*. 2021;24(2):160-67.
- 54 Van Overwalle F. Social and emotional learning in the cerebellum. *Nat Rev Neurosci*. 2024;25(12):776-91.
- 55 Ashida R, Cerminara NL, Edwards RJ, Apps R, Brooks JCW. Sensorimotor, language, and working memory representation within the human cerebellum. *Hum Brain Mapp*. 2019;40(16):4732-47.
- 56 Moberget T, Ivry RB. Prediction, Psychosis, and the Cerebellum. *Biol Psychiatry Cogn Neurosci Neuroimaging*. 2019;4(9):820-31.
- 57 Tonna M, Ottoni R, Ossola P, De Panfilis C, Marchesi C. Late-onset obsessive-compulsive disorder associated with left cerebellar lesion. *Cerebellum*. 2014;13(4):531-5.
- 58 Yang Z, Xiao S, Su T, Gong J, Qi Z, Chen G, et al. A multimodal meta-analysis of regional functional and structural brain abnormalities in obsessive-compulsive disorder. *European archives of psychiatry and clinical neuroscience*. 2024;274(1):165-80.
- 59 Tao Q, Dang J, Niu X, Gao X, Zhang M, Yang Z, et al. White matter microstructural

- abnormalities and gray matter volume alterations in obsessive-compulsive disorder: A coordinate-based meta-analysis. *Journal of affective disorders*. 2023;320:751-61.
- 60 Li H, Wang Y, Xi H, Zhang J, Zhao M, Jia X. Alterations of regional spontaneous brain activity in obsessive-compulsive disorders: A meta-analysis. *Journal of psychiatric research*. 2023;165:325-35.
- 61 Tanaka H, Ishikawa T, Lee J, Kakei S. The Cerebro-Cerebellum as a Locus of Forward Model: A Review. *Front Syst Neurosci*. 2020;14:19.
- 62 Popa LS, Ebner TJ. Cerebellum, Predictions and Errors. *Front Cell Neurosci*. 2018;12:524.



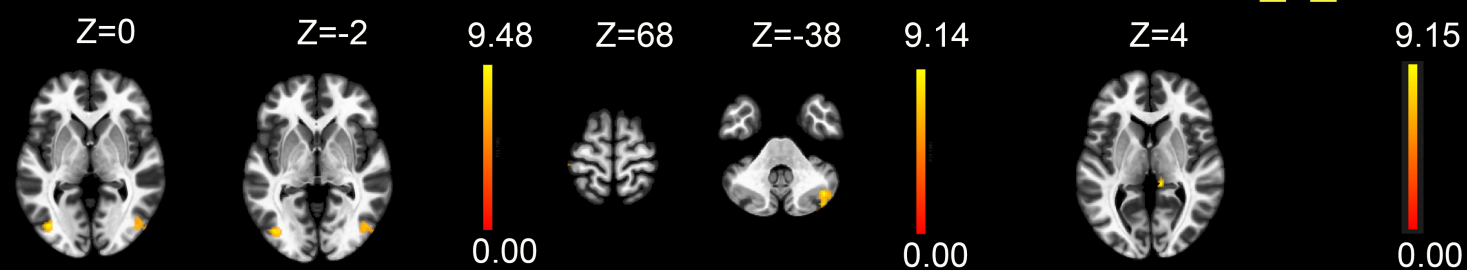
SomMotA_1_l

Thalamus_l



SomMotA_2_l

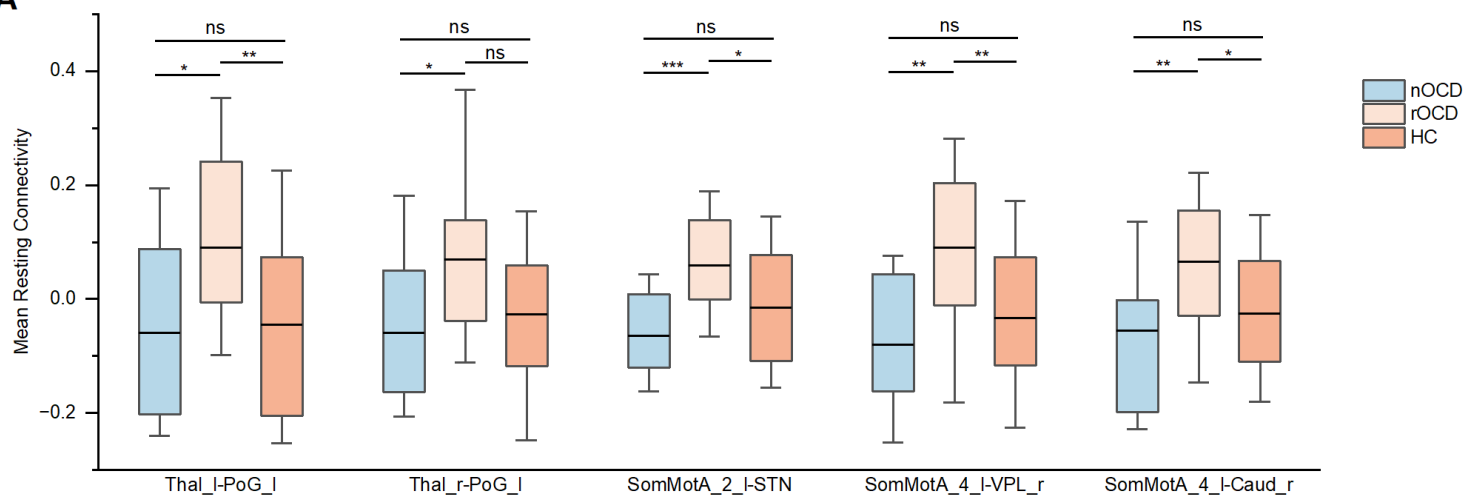
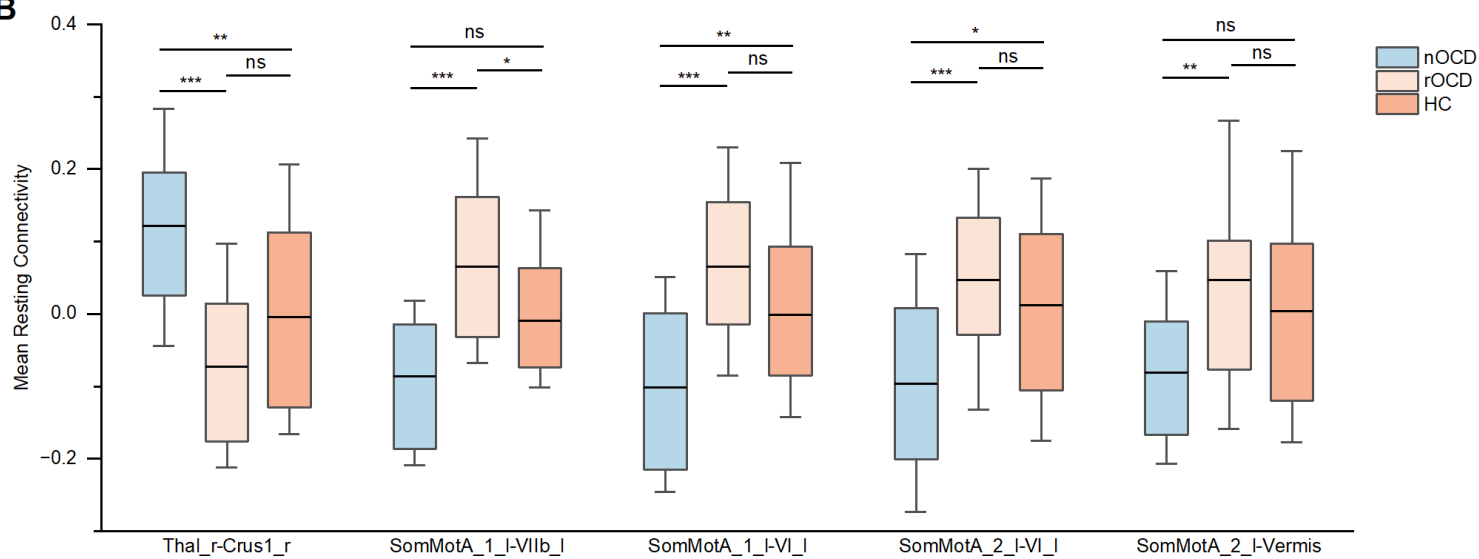
SomMotA_4_l

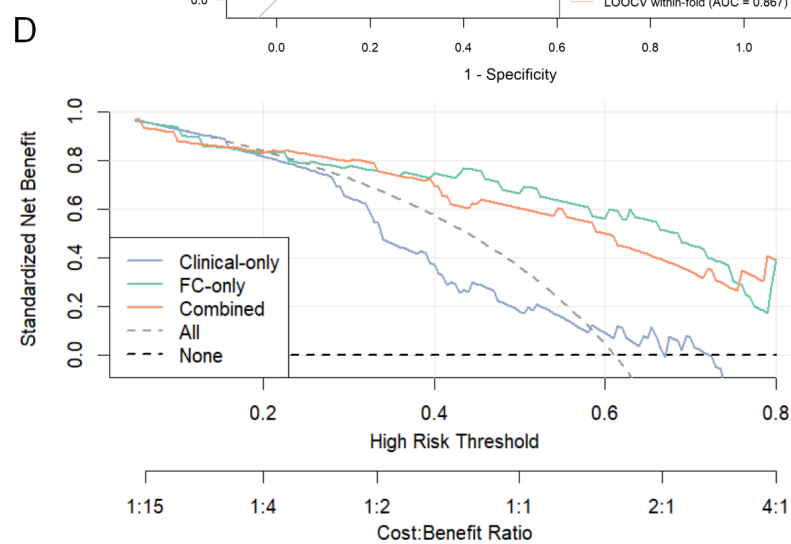
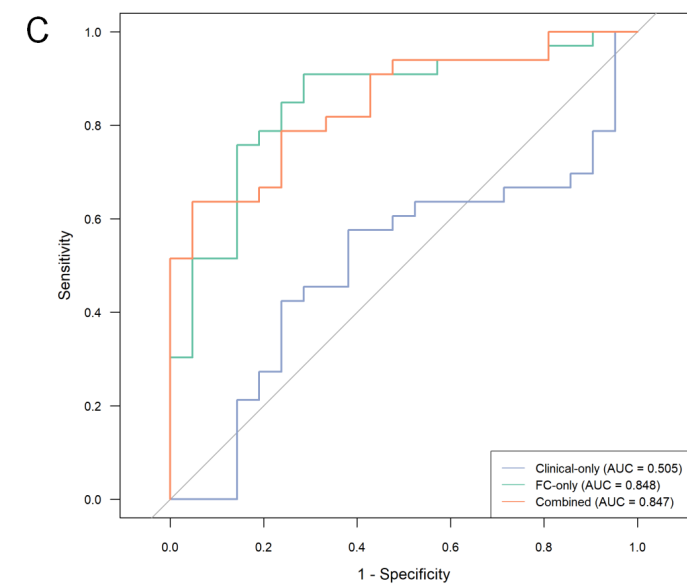
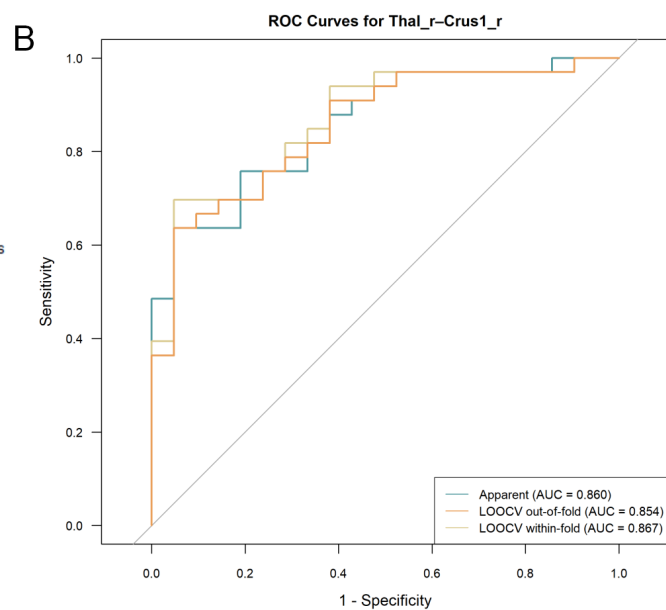
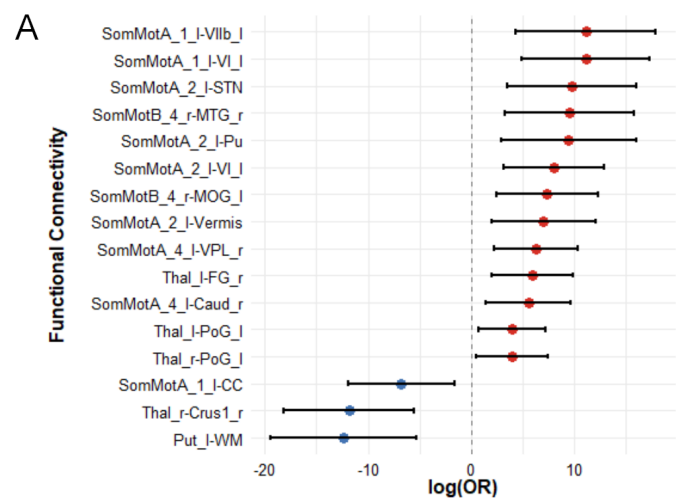


SomMotB_4_r

Thalamus_r

Putamen_l

A**B**



Supplementary Table 11. Cross-validated analyses under different cut-offs of Y-BOCS score reduction.

Response Threshold	FC	Responders	Non-responders	AUC_cv	95%CI_cv_low
0.25	Thal_r-Crus1_r	39	15	0.687	0.539
0.25	Thal_r-PoG_l	39	15	0.523	0.313
0.25	Put_l-WM	39	15	0.682	0.502
0.25	Thal_l-FG_r	39	15	0.615	0.418
0.25	Thal_l-PoG_l	39	15	0.508	0.306
0.25	SomMotA_1_l-VIIb_l	39	15	0.607	0.427
0.25	SomMotA_1_l-VI_l	39	15	0.562	0.372
0.25	SomMotA_1_l-CC	39	15	0.590	0.386
0.25	SomMotA_2_l-STN	39	15	0.602	0.421
0.25	SomMotA_2_l-VI_l	39	15	0.549	0.338
0.25	SomMotA_2_l-Vermis	39	15	0.566	0.379
0.25	SomMotA_2_l-Pu	39	15	0.573	0.389
0.25	SomMotA_4_l-VPL_r	39	15	0.561	0.365
0.25	SomMotA_4_l-Caud_r	39	15	0.585	0.403
0.25	SomMotB_4_r-MTG_r	39	15	0.568	0.352
0.25	SomMotB_4_r-MOG_l	39	15	0.528	0.304
0.3	Thal_r-Crus1_r	35	19	0.794	0.676
0.3	Thal_r-PoG_l	35	19	0.561	0.382
0.3	Put_l-WM	35	19	0.693	0.543
0.3	Thal_l-FG_r	35	19	0.606	0.432
0.3	Thal_l-PoG_l	35	19	0.576	0.399
0.3	SomMotA_1_l-VIIb_l	35	19	0.720	0.567
0.3	SomMotA_1_l-VI_l	35	19	0.681	0.517
0.3	SomMotA_1_l-CC	35	19	0.571	0.387
0.3	SomMotA_2_l-STN	35	19	0.624	0.461
0.3	SomMotA_2_l-VI_l	35	19	0.635	0.459
0.3	SomMotA_2_l-Vermis	35	19	0.636	0.469
0.3	SomMotA_2_l-Pu	35	19	0.611	0.444
0.3	SomMotA_4_l-VPL_r	35	19	0.627	0.450
0.3	SomMotA_4_l-Caud_r	35	19	0.558	0.382
0.3	SomMotB_4_r-MTG_r	35	19	0.651	0.484
0.3	SomMotB_4_r-MOG_l	35	19	0.618	0.448
0.5	Thal_r-Crus1_r	21	33	0.654	0.505
0.5	Thal_r-PoG_l	21	33	0.560	0.395
0.5	Put_l-WM	21	33	0.596	0.436
0.5	Thal_l-FG_r	21	33	0.652	0.499
0.5	Thal_l-PoG_l	21	33	0.563	0.401
0.5	SomMotA_1_l-VIIb_l	21	33	0.658	0.509
0.5	SomMotA_1_l-VI_l	21	33	0.696	0.552
0.5	SomMotA_1_l-CC	21	33	0.592	0.436
0.5	SomMotA_2_l-STN	21	33	0.586	0.428
0.5	SomMotA_2_l-VI_l	21	33	0.683	0.535

0.5	SomMotA_2_l-Vermis	21	33	0.610	0.453
0.5	SomMotA_2_l-Pu	21	33	0.592	0.434
0.5	SomMotA_4_l-VPL_r	21	33	0.567	0.402
0.5	SomMotA_4_l-Caud_r	21	33	0.632	0.477
0.5	SomMotB_4_r-MTG_r	21	33	0.597	0.442
0.5	SomMotB_4_r-MOG_l	21	33	0.683	0.539

Threshold-based indices derived from the Youden index in the training set. AUC, the odds ratio; CI, confidence interval; Thal, thalamus; Crus1, cerebellar Crus I; PoG, postcentral gyrus; Put, putamen; WM, white matter; FG, fusiform gyrus; cerebellar lobule VI; CC, corpus callosum; STN, subthalamic nucleus; Vermis, cerebellar vermis VII; Pu, pulvinar; Temporal Gyrus; MOG, Middle Occipital Gyrus; l, left; r, right.

95%CI_cv_high	Optimal FC threshold	Sensitivity	Specificity	<i>p</i> -value (FDR-adjusted)
0.835	-0.044	0.641	1.000	0.159
0.733	-0.151	0.923	0.400	0.495
0.862	0.008	0.744	0.933	0.159
0.813	-0.068	0.821	0.733	0.366
0.710	-0.176	0.872	0.467	0.537
0.786	-0.044	0.692	0.733	0.366
0.753	-0.047	0.744	0.667	0.366
0.793	0.113	0.872	0.400	0.845
0.783	0.028	0.538	0.933	0.366
0.759	-0.021	0.667	0.667	0.758
0.753	-0.141	0.897	0.533	0.366
0.757	0.044	0.513	0.867	0.366
0.756	0.102	0.410	1.000	0.366
0.766	-0.049	0.795	0.667	0.366
0.783	-0.034	0.718	0.733	0.366
0.752	-0.125	0.923	0.400	0.495
0.912	-0.050	0.657	0.947	0.002
0.740	-0.116	0.914	0.421	0.247
0.844	0.000	0.743	0.842	0.052
0.780	0.005	0.686	0.737	0.138
0.753	-0.165	0.886	0.474	0.227
0.874	-0.035	0.686	0.737	0.029
0.846	-0.047	0.771	0.632	0.058
0.756	0.109	0.886	0.474	0.227
0.787	0.028	0.543	0.842	0.123
0.810	0.011	0.629	0.789	0.123
0.803	-0.116	0.886	0.526	0.123
0.777	0.052	0.486	0.842	0.137
0.805	0.055	0.543	0.842	0.123
0.734	-0.030	0.714	0.684	0.247
0.819	-0.034	0.800	0.789	0.112
0.788	-0.059	0.771	0.632	0.127
0.802	0.087	0.952	0.424	0.083
0.725	-0.049	0.667	0.545	0.235
0.756	-0.026	0.667	0.667	0.177
0.805	0.009	0.714	0.606	0.083
0.724	-0.165	0.905	0.333	0.235
0.807	0.028	0.619	0.788	0.083
0.839	-0.038	0.810	0.606	0.065
0.747	0.101	0.905	0.364	0.177
0.743	0.075	0.429	0.818	0.183
0.830	-0.014	0.762	0.576	0.065

0.768	-0.038	0.762	0.545	0.177
0.749	0.088	0.381	0.909	0.177
0.732	0.102	0.476	0.818	0.235
0.787	-0.030	0.762	0.545	0.122
0.752	-0.034	0.857	0.576	0.177
0.826	0.195	0.333	0.970	0.065

interval; area under the curve; cv, cross-validation; FDR, false discovery rate; gyrus; SomMot, somatomotor network; VIIb, cerebellar lobule VIIb; VI, r; VPL, ventral posterolateral nucleus; Caud, caudate; MTG, Middle